

■ Lalitha's Math Academy

Comprehensive System Documentation & Technical Delivery Guide

1. ■ Executive Summary

Lalitha's Math Academy Tuition Platform is a full-stack web application designed for managing student registrations, class scheduling, custom booking packages, and automated admin management. The platform uses **Google Sheets** as its real-time database, enabling direct spreadsheet access for the academy administrator (V. Lalitha Ma'am) without complex database overhead.

2. ■ Admin Authentication & Credentials

Admin Portal URL	/admin-page (e.g., https://your-domain.com/admin-page)
Admin Password	Lalitha@MathAcademy
Security Enforcement	Strict single-password check with SHA-256 hash backup in Google Sheets admin worksheet.

3. ■■ Technology Stack Architecture

Component	Technology	Description
Backend Framework	Flask (Python 3.11)	Python web framework managing session auth, API endpoints, and dynamic HTML rendering.
Database Integration	GSpread / GSheets API v4	Real-time spreadsheet read/write with automatic service account authentication.
Frontend UI / UX	HTML5 / Vanilla CSS3 / JS	Modern glassmorphism UI, responsive layouts, interactive day/time pickers.
WSGI Server	Gunicorn 22.0	Production-grade WSGI HTTP server for Render cloud deployment.
Cloud Hosting	Render Free Plan	Auto-deploy container with Procfile & render.yaml configurations.
Notifications	WhatsApp wa.me & CallMeBot	Direct pre-filled WhatsApp links & automatic status notifications.

4. ■ Detailed Module-by-Module Working

A. ■ Student Registration & Authentication

- **Registration Form:** Collects student's Full Name, Class (Grade 1 to 12), School Name, Address, Pincode, 10-digit Phone Number, and Password. Creates a new record in GSheets users sheet.
- **Login Form:** Validates 10-digit phone number and SHA-256 hashed password. Creates secure session for student dashboard access.

B. ■ Student Booking Engine (4 Flexible Plans)

- **1. Hourly Plan (hourly):** Student specifies **Hours Needed** (1–4 hrs), selects an available time slot from live GSheets data, picks a preferred start date, and chooses class mode (Offline/Online). Day picker is hidden for hourly.
- **2. Weekly Plan (weekly):** Student enters number of days per week, selects specific days (including Sunday), selects hours per session, chooses an available time slot, and sets total duration in weeks.
- **3. Monthly Package (monthly):** Offers a ■ **Daily (Mon–Sun)** toggle button or **Specific Days/Week** option. Student sets number of months, hours per session, time slot from GSheets, and preferred start date.

- **4. Yearly Program (yearly):** Designed for long-term board exam prep (3, 6, 9, 12 months). Supports **Daily** or **Specific Days** schedule, hours per class, and GSheets time slots.

C. Admin Management Portal (Password: Lalitha@MathAcademy)

- **Overview Dashboard:** Real-time counters for Pending Requests, Confirmed Sessions, and Slot Availability.
- **Pending Requests Tab:** Lists all pending bookings with student phone, plan, days, time slot, and mode. Admin can enter Google Meet links for online classes and click ☐ Confirm or ☐ Remind.
- **Confirmed Sessions Tab:** Displays active confirmed sessions with student contact info, meet links, and direct WhatsApp buttons.
- **Availability Slot Manager:** Interactive toggle switches for every day of the week (Monday through **Sunday**). Clicking ☐ Save Availability updates GSheets in real-time.
- **Academy Settings:** Allows admin to customize available days, max days per week, default time slots list, min/max hours, and pricing display hints.
- **Registered Students Directory:** View complete student database with school name, address, pincode, and registration timestamp.

5. ■ Google Sheets Database Structure (9 Worksheets)

Worksheet Name	Key Columns / Schema	Purpose
users	id, full_name, class, school, address, pincode, phone, password_hash, created_at	Student accounts & login credentials.
admin	username, password_hash	Admin account hash.
settings	key, value, description	Academy preferences & default slots.
bookings	id, user_id, full_name, phone, class, school, mode, basis, quantity, days_of_week, time_slot, preferred_date, status, meet_link	All booking requests.
confirmed_bookings	(Same schema as bookings)	Confirmed sessions index.
available_slots	day, time_slot, is_available	Live availability toggles (Mon–Sun).
carousel_images	url, caption, order	Homepage hero carousel images.
pricing	basis, description, price_hint, highlight	Plan features & pricing hints.
contact_info	key, value	Phone, email, address & map embed.

6. ■ Render Free Plan Deployment Guide

To deploy this platform on Render Free Web Service:

1. Push repository to GitHub.
2. Create a new **Web Service** on Render connected to your repository.
3. Render automatically detects Procfile and render.yaml.
4. In Render Dashboard Environment Variables, set:

```
PORT = 10000
ADMIN_PASSWORD = Lalitha@MathAcademy
GOOGLE_SHEET_ID = your_spreadsheet_id
GOOGLE_CREDENTIALS_JSON = paste_JSON_key_contents
```